

ScotRec: Social Chain-of-Thought LLM Reasoning for Recommendation

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Abstract

Large language models (LLMs) have emerged as a promising paradigm for recommender systems, due to their powerful capabilities in global knowledge integration and reasoning. However, LLMs are inherently prone to confirmation bias – the tendency to favor information that reinforces users’ existing views – which leads to an overemphasis on previously shown viewpoints and ignores diverse user beliefs for recommendations. To address this issue, in this paper, we propose SCoTRec, a social chain-of-thought reasoning framework for recommendation. SCoTRec first constructs sentiment-aware user profiles by extracting sentiment terms from user reviews. It then incorporates users’ social sentiment information into the social chain-of-thought reasoning units to improve recommendations. In particular, we categorize the social chain-of-thought into sentiment-based pathways and apply human evaluation operations – backtracking, discarding, retaining, and aggregating – to simulate nuanced sentiment cognition and interpersonal influence, effectively alleviating confirmation bias. Extensive experiments on four benchmark datasets demonstrate the effectiveness of SCoTRec in alleviating confirmation bias and improving recommendations.

CCS Concepts

• Information systems → Recommender systems.

Keywords

Social Networks, Large Language Model, Chain of Thought, Recommender Systems.

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1 Introduction

Recommender systems (RSs) [3, 21, 23, 24, 52, 62] play a vital role in delivering relevant information across content-rich platforms such as social media, music streaming services, and digital libraries. Among various approaches in RSs, collaborative filtering (CF) [2, 25, 32, 37] stands out as a fundamental technique, which learns user and item representations from the user-item interaction matrix to infer user preferences. Despite its widespread adoption, CF struggles with the cold start problem, primarily due to the sparsity of historical interaction data.

With the rapid development of Large Language Models (LLMs), LLMs for recommendation (LLM4Rec) [13, 17, 31, 48] leverages the exceptional world knowledge and reasoning capabilities of LLMs, offering a viable solution to the cold start problem. Early studies of LLM4Rec attempt to incorporate user-item collaborative information into LLMs. For instance, researchers attempt to integrate collaborative embeddings from conventional recommendation models (e.g., matrix factorization [22, 35], collaborative filtering [19, 42]) into the LLMs’ embedding space [56, 57]. However, they still suffer from a major challenge for effective knowledge transfer, due to the significant differences in knowledge distribution and semantic spaces between LLMs and these traditional recommendation models. Subsequently, the LLM4Rec methods introduce a reinforcement learning mechanism based on human feedback (RLHF) [8, 36, 50] to align user behavior with LLMs. Although effective to some extent, these LLM4Rec methods face the limitation of high computational demands [10]. To address this issue, LLM role-playing [44, 49] has emerged. It leverages user profiles to generate interpretable, human-aligned content in order to enhance recommendation performance.

Although LLM role-playing methods humanize the “algorithm”, most existing methods for recommendation do not perceive the user’s sentiment. Moreover, most existing LLM role-playing methods with user profiles overlook the susceptibility of LLMs to confirmation bias [15, 20, 38]. That is, the feedback loop in LLMs tends to prioritize content that reinforces users’ existing viewpoints, which

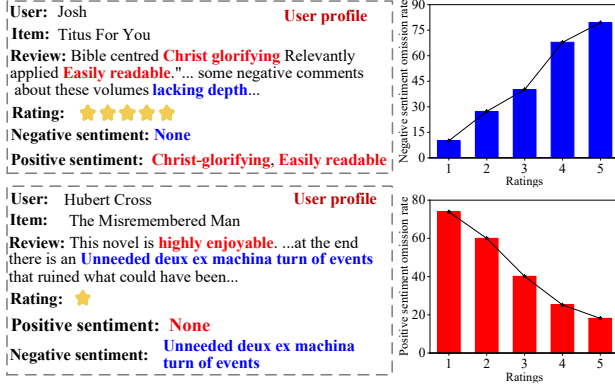


Figure 1: Illustration of confirmation bias in LLM4Rec.

leads to an overemphasis on previously shown viewpoints and ignores diverse user beliefs for recommendations. For example, as shown in Figure 1, when we ask the LLM (i.e., LLaMA 3.1-8B [43]) to take user subjective ratings into account and extract both positive and negative sentiment factors from user reviews, the model identifies only one type of sentiment (either positive or negative) rather than both. This indicates that the LLM tends to rely on the given rating and automatically overlooks different user perspectives toward the same item. We then conducted a further analysis of the LLM’s sentiment omission rates across different rating levels. The results reveal a clear pattern: as ratings become higher (e.g., rating = 5), the model increasingly ignores negative sentiment factors and favors positive sentiment (high blue bars vs. low red bars); conversely, for lower ratings (e.g., rating = 1), it is more likely to overlook positive sentiment and favor negative sentiment (low blue bars vs. high red bars). This suggests that the feedback loop in LLM4Rec amplifies the effect of confirmation bias (favoring positive or negative), leading to an overestimation of users’ viewpoints and degrading recommendation performance.

To address these issues, we propose an innovative LLM role-playing method, which integrates the Social Chain-of-Thought (CoT) paradigm for Recommendation, called SCoTRec. SCoTRec mainly consists of two stages: sentiment-aware user profile construction and social CoT reasoning. Given that the reward system of the human brain is highly dependent on interactions with others and providing sentiment feedback [59], we first construct a sentiment-aware user profile by extracting sentiments from user reviews. Then, we introduce users’ social network and propose a social CoT framework to incorporate social sentiment feedback into each reasoning unit. Specifically, we divide the social CoT reasoning into sentiment-based paths (e.g., positive-positive, negative-positive, negative-negative) and implement operations such as “backtracking”, “discarding”, “retaining”, and “aggregating” to simulate human-like sentiment cognition and interpersonal influence. By modeling diverse user beliefs based on the sentiments of users’ social networks, SCoTRec effectively alleviates the confirmation bias in the LLM feedback loop. We summarize the main contributions of this paper as follows:

- We observe the confirmation bias in the LLM4Rec feedback loop and provide an effective solution to address it.

- We propose a novel LLM4Rec method, SCoTRec, which builds sentiment-aware user profiles and introduces a pioneering social CoT mechanism that integrates social sentiment feedback into the individual reasoning process.
- We conduct empirical experiments on four public recommendation datasets, demonstrating the effectiveness of SCoTRec in alleviating confirmation bias and improving recommendations.

2 Preliminary

2.1 Review-based LLM for Recommendation

Given a user set $\mathcal{U}$ with N_u users and an item set $\mathcal{V}$ with N_v items, the user-item interaction is represented by the rating matrix $R \in \mathbb{R}^{N_u \times N_v}$. The $r_{u,v} \in R$ (e.g., $r=1, 2, 3, 4, 5$) denotes the explicit rating given by user u to item v . Concurrently, the review set is $\mathcal{T} = \mathbb{R}^{N_u \times N_v}$, where each element $t_{u,v} \in \mathcal{T}$ represents the review of user u on item v . The primary task is to predict the user ratings [18, 44, 51, 58], where the LLM (f_{LLM}) learns the mapping to predict unobserved ratings $\hat{r}_{ij}$ for items that have not yet reviewed as accurately as possible:

$$f_{LLM} : (u, v, t_{u,v}) \rightarrow \hat{r}_{u,v}. \quad (1)$$

The predicted ratings are then used for scores to recommend items to the user.

2.2 Chain of Thought Reasoning

CoT reasoning [5, 47, 53, 55] is a technique that prompts LLMs to generate a series of intermediate reasoning units before arriving at the final answer. Given an input x , the intermediate thought units $z_1, z_2, \dots, z_n$ are generated sequentially, with each z conditioned on the input x and all previous thoughts. This process is expressed as:

$$z_{n+1} = \arg \max_z p_\theta(z|x, z_1, \dots, z_n). \quad (2)$$

Finally, the output y is derived from the entire sequence of thoughts, conditioned on the input x and all the reasoning units $z_1, z_2, \dots, z_{n+1}$. This is expressed as:

$$y = \arg \max_y p_\theta(y|x, z_1, \dots, z_{n+1}). \quad (3)$$

Where the output distribution of an LLM parameterized by θ is denoted by p_θ . This formulation shows that the reasoning process involves a series of reasoning units where the intermediate thoughts are generated in a sequential manner, and the final output y is determined by maximizing the conditional probability based on the complete reasoning path.

3 Methodology

In this section, we present the proposed method SCoTRec, which consists of three main modules: 1) sentiment-aware modeling, 2) user profile construction, and 3) social CoT reasoning. The first module focuses on modeling sentiment from user reviews, and the second focuses on constructing user profiles. The third module simulates human evaluation and interpersonal influence to reason the positive and negative traits of items for recommendations. The overall framework is illustrated in Figure 2.

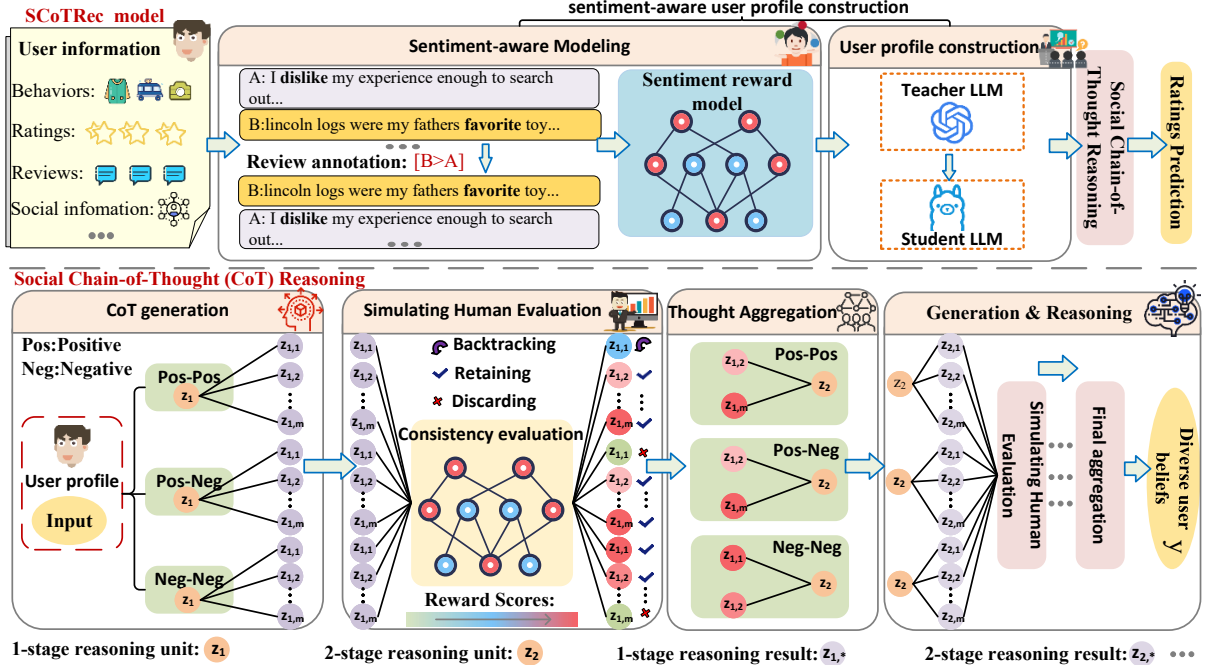


Figure 2: The illustration of the proposed SCoTRec framework.

3.1 Sentiment-aware Modeling

Traditional pre-trained sentiment-aware models suffer from the lack of relative preference information, making them ineffective in accurately modeling sentiment perception. These scalar ratings typically provide a vague evaluation of quality and are inadequate for capturing subjective phenomena such as sentiment, where pre-trained models often exhibit significant variance in their judgments [36].

To overcome this limitation, we randomly select a subset of reviews and, for each review, generate a sequence containing N corresponding reviews. For example, for a review t , its associated sequence is represented as $S_{review}^t = \{t_1, t_2, \dots, t_N\}$. Each review within the sequence are then manually ranked according to the perceived strength of user sentiment, ordered from positive to negative, as follows:

$$t_{\pi_1} \succ t_{\pi_2} \succ \dots \succ t_{\pi_N}, \quad (4)$$

where π denotes the ranking order and $t_{\pi_{k+1}} \succ t_{\pi_k}$ indicates that comment $t_{\pi_{k+1}}$ is judged to express a more positive sentiment than t_{π_k} . And these ranked review sequences S_{rank} are obtained, S_{rank}^t represents the ranked review sequence of review t .

We utilize these ranked review sequences to fine-tune a sentiment-aware reward model, thereby transforming the collected relative sentiment preferences into scalar reward signals. The reward model is instantiated with a pre-trained language model (Llama-3.1-8B-Instruct). For a given ranked review sequence S_{rank}^t , the reward model can be formalized as follows:

$$\text{reward}_\theta(t_{\pi_k}) = \text{MLP}_\phi(\text{Encoder}_\phi(t_{\pi_k})), \quad (5)$$

where $\text{reward}_\theta(t_{\pi_k})$ is the sentiment score assigned to review t_{π_k} ($t_{\pi_k} \in S_{rank}^t$). Encoder $_\phi$ denotes a pre-trained language model employed as a feature extractor, and MLP_ϕ is a linear regression head. The full parameter set $\theta = \{\phi, \phi\}$ comprises the parameters for the encoder and the regressor, respectively. To align this reward model with human sentiment preferences, we minimize the following pairwise ranking loss function over all ranked sets (representing items $\mathcal{V}$) in our annotated dataset:

$$\mathcal{L} = -\frac{1}{\mathcal{Z}} \sum_{1 \leq i < j \leq N} \log \sigma(\text{reward}_\theta(t_{\pi_i}) - \text{reward}_\theta(t_{\pi_j})), \quad (6)$$

where $\mathcal{Z} = \frac{N(N-1)}{2}$ represents the total number of unique pairs across the ranked review sequence S_{rank}^t , and $\sigma(\cdot)$ is the sigmoid function. Finally, the fine-tuned sentiment reward model labels reviews (positive or negative).

3.2 User Profile Construction

Teacher-student model. To fully leverage GPT-4o's powerful reasoning capabilities and reduce computational costs, we adopt the teacher-student model (GPT-4o) in both the user profile construction and social CoT reasoning modules, where the distilled student models (LLaMA3.1-8B) are applied to user profile construction and the 3.3 social CoT reasoning unit, respectively. The teacher model is defined as follows:

$$y_{u,v}^{GPT} = f_{gpt}(x_{u,v}), \quad (7)$$

where $x_{u,v}$ is the constructed prompt based on $t_{u,v}$, and $y_{u,v}^{GPT}$ is the corresponding structured output from GPT-4o. The teacher-generated data serves as supervision for distilling the student summarization model.

Given teacher-generated data $\mathcal{D} = \{(x_{ij}, y_{ij}^{GPT}) | t_{ij} \in \mathcal{T}\}$, we fine-tune a student model based on the LLaMA architecture, aiming to distill the teacher into a lighter and more efficient model. Using QLoRA [9], we fine-tune the LLaMA-based student to replicate the teacher's outputs under a standard supervised instruction-tuning setup. Specifically, the student model receives the same prompt x_{ij} and is optimized to autoregressively generate the corresponding target content y_{ij}^{GPT} . The training loss is defined as the negative log-likelihood of the ground-truth tokens:

$$\mathcal{L}_{gen} = \mathbb{E}_{(x,y) \sim \mathcal{D}} [-\log p_{\theta}(y|x)], \quad (8)$$

where p_{θ} represents the output distribution of the LLM parameterized by θ and θ denotes the trainable parameters of the student LLaMA model within the QLoRA.

User Profile Construction. For each item, we extract positive and negative aspect-level sentiment terms, with the corresponding prompt template provided in the appendix A.2. We then construct user profiles that incorporate these sentiment details. In this way, users' sentiment reactions to items can be expressed using detailed, aspect-level positive and negative terms. The profile of user u on item v is expressed as:

$$\mathcal{E}_{user}^{u,v} = (E_{pos}, E_{neg}, \mathcal{H}_v), \quad (9)$$

where E_{pos} and E_{neg} represent positive and negative sentiment terms, respectively, and $\mathcal{H}_v$ represents the user's contexts (review, rating) for item v .

3.3 Social CoT Reasoning

LLMs tend to prioritize information consistent with users' existing viewpoints. To address this issue, we propose social CoT, which integrates aspect-level sentiment terms in user social networks into CoT to mitigate confirmation bias.

CoT Generation. We denote the profiles of users who have a social relationship with user u and have interacted with item v as $\mathcal{E}_{other}^{u,v}$. The overall social network profile is then defined as $\mathcal{E}_{social}^{u,v} = (\mathcal{E}_{user}^{u,v}, \mathcal{E}_{other}^{u,v})$. We incorporate the user's social network profile $\mathcal{E}_{social}^{u,v}$ into CoT reasoning to alleviate confirmation bias. Based on the sentiment label of user reviews from the "3.1 Sentiment-aware Modeling", we classify the reasoning chains into three sentiment categories $\mathcal{S}_{chain}$: "Pos-Pos", "Pos-Neg", and "Neg-Neg", where Pos-Pos chains group user profiles with positive review labels in the reason process, Neg-Neg chains group those with negative review labels, and Pos-Neg chains group positive and negative review labels. Given the user profile $\mathcal{E}_{user}^{u,v}$, for reasoning unit z_i , the reasoning results $\{z_{i,1}, z_{i,2}, \dots, z_{i,m}\}$ are generated in parallel, conditioned on both the user's social network profile $\mathcal{E}_{social}^{u,v}$ and the previous round reasoning results. The entire reasoning process can be expressed as:

$$\{z_{i,1}^{\delta}, \dots, z_{i,m}^{\delta}\} = \arg \max_z p_{\theta}(z | \mathcal{E}_{social}^{u,v}, z_{i-1,1}^{\delta}, \dots, z_{i-1,m}^{\delta}), \delta \in \mathcal{S}_{chain}, \quad (10)$$

where $z_{i,m}^{\delta}$ represents the m -th reasoning result of z_i^{δ} reasoning unit and δ represents sentiment categories hyperparameters. Finally, the reasoning output (diverse user beliefs) $y_{u,v}$ is derived from the

reasoning chain and can be expressed as:

$$y_{u,v}^{\delta} = \arg \max_y p_{\theta}(y | \mathcal{E}_{social}^{u,v}, z_1^{\delta}, \dots, z_{n+1}^{\delta}), \delta \in \mathcal{S}_{chain} \quad (11)$$

The final output of a given reasoning chain, $y_{u,v}^{\delta}$, is determined by maximizing the conditional probability along the entire reasoning path, and p_{θ} denotes the output distribution of the LLM parameterized by θ . The CoT reasoning prompt is as follows:

CoT Reasoning Prompt
<Instruction> Here is information about a user and a new {product_class} "{target_title}" being recommended to the user. For the user, we have the user's review history. For the new item being recommended, we have the item review history by trusted users (social network). Analyze whether the user will like the new {product_class} "{target_title}". Give you a rationale in one paragraph.
<User profile>
{user_history_text}
<User's social network profile>
{social_network_history_text}

Simulating Human Evaluation Operations. In the social CoT, to simulate human evaluation operations, we summarize human evaluation as "backtracking", "discarding", "retaining", and "aggregating". We first define the scoring function to decide which operation of "backtracking", "discarding", and "retaining" to perform. To avoid contradictory or low-quality information and prevent error propagation in the reasoning chain, we introduce a consistency evaluation score function that evaluates the quality of the current reasoning result based on its semantic consistency [14] with the previous reasoning fragments. The decision process is governed by the following conditions:

$$\begin{cases} \text{Backtrack}(z_{i,j-1}^{\delta}) & \text{if } \frac{1}{s} \sum_{j=1}^s \text{Score}(z_{i-1,j}^{\delta}) < \text{Score}(z_{i,j-1}^{\delta}) < \epsilon, \\ \text{Discard}(z_{i,j-1}^{\delta}) & \text{if } \text{Score}(z_{i,j-1}^{\delta}) < \frac{1}{s} \sum_{j=1}^s \text{Score}(z_{i-1,j}^{\delta}), \\ \text{Retain}(z_{i,j-1}^{\delta}) & \text{if } \text{Score}(z_{i,j-1}^{\delta}) \geq \epsilon, \end{cases} \quad (12)$$

where ϵ represents the threshold hyperparameter, and $\text{Score}(z_{i-1,j})$ is the consistency evaluation score function, which evaluates the quality of the current reasoning result by counting the number of information fragments that are consistent with those from the previous round. The information fragments are divided as follows:

$$z_{i,j}^{\delta} = \{\gamma_{i,j}^{(1)}, \gamma_{i,j}^{(2)}, \dots, \gamma_{i,j}^{(s)}\}, \quad (13)$$

where $\gamma_{i,j}^{(p)}$ represents the p -th information fragment within the reasoning result $z_{i,j}^{\delta}$. The information fragment usually corresponds to the smallest comparable fact or property description, e.g., "The food is tasty but the drinks are weak." Split into $\gamma^{(1)}$: food is tasty, $\gamma^{(2)}$: drinks are weak. In the first round, the user review is compared. The functions *Backtrack*($\cdot$), *Discard*($\cdot$), and *Retain*($\cdot$) correspond to the operations of backtracking (re-requesting), discarding, and retaining, respectively. The maximum number of backtracks is 3; beyond that, the result is discarded. After the evaluation is completed, we aggregate the retained reasoning results, where only the backtracked results judged to be retained will be included.

Thought Aggregation. The goal of aggregation is to combine the reasoning results by weighing their contributions according to the scores. We adopt the dominant-candidate-first strategy and incremental merging. First, we sort the reasoning results by their consistency evaluation scores, from highest to lowest $\{z_{\pi_1}^\delta, z_{\pi_2}^\delta, \dots, z_{\pi_l}^\delta\}$, where π denotes the ranking order, and $z_{\pi_1}^\delta$ represents the highest-scoring reasoning result, with the remaining results following in descending order accordingly. In the aggregation phase, the one with the highest score is first selected as the candidate reasoning result. Then, for other reasoning results, only the information fragments not included in the candidate reasoning results are extracted and incrementally merged into the aggregation set, which can be formalized as:

$$\hat{z}_{i+1}^\delta = z_{i,\pi_1}^\delta \oplus z_{i,\pi_2}^\delta \oplus \dots \oplus z_{i,\pi_l}^\delta, \quad (14)$$

where $\hat{z}_{i+1}^\delta$ serves as the “{user_history_text}” of the z_{i+1}^δ reasoning unit prompt template, z_{i,π_1}^δ represents the reason result with the largest score generated by reason unit z_i^δ and $\oplus$ is the order-preserving and duplicate-removing concatenation operator [63], which is defined as follows:

$$z_{i,\pi_1}^\delta \oplus z_{i,\pi_2}^\delta = \{\gamma | \gamma \in z_{i,\pi_1}^\delta\} || \{\gamma | \gamma \in z_{i,\pi_2}^\delta, \gamma \notin z_{i,\pi_1}^\delta\}, \quad (15)$$

where γ represents information fragment, $||$ denotes sequence concatenation. In words, the operator first retains all elements of z_{i,π_1}^δ in their original order, and then iterates over z_{i,π_2}^δ to append only those elements that do not already appear in z_{i,π_1}^δ .

Generation & Reasoning. We use the aggregated result as the “{user_history_text}” for the next round of reasoning units. Finally, we aggregate the “Pos-Pos,” “Pos-Neg,” and “Neg-Neg” reasoning chains to obtain the final reasoning output $y_{u,v}$ of social CoT reasoning, guided by the final aggregation prompt. The prompt template is given below:

Final Aggregation Prompt
<Instruction> The following are the speculations and reasons for whether users like new items under different emotions, please analyze their reasonableness and merge the three paragraphs into one paragraph. Give you rationale in one paragraph. <Speculation and Reason> {input1} {input2} {input3}

Through social CoT reasoning, we derive the diverse user beliefs $y_{u,v}$, which contain interpersonal sentiment traits—that is, the positive and negative sentiment traits of the user and their social network toward the item. We then incorporate the diverse user beliefs $y_{u,v}$ into the recommendation model for rating prediction, which enables the model to determine whether the item should be recommended to the user.

$$\hat{r}_{u,v} = f_{LLM}(u, v, t_{u,v}, y_{u,v}), \quad (16)$$

where $\hat{r}_{u,v}$ represents the final prediction rating of user u for item v , and $f_{LLM}(\cdot)$ is the fine-tuned LLM (LLaMA3.1-8B) for rating prediction.

4 Experiments

4.1 Datasets

We conducted experiments on four datasets: Epinions and Yelp as social datasets, and Music and Book as non-social ones. For datasets lacking explicit social links, we established implicit user connections by linking users who interacted with the same items. Following prior work [12, 18], each dataset was split into training, validation, and test sets in an 8:1:1 ratio. For Yelp, only data from the most recent six months were used. To ensure fair evaluation, cold-start users and items—those absent from the training set—were excluded from the validation and test sets, consistent with previous studies [12, 18].

4.2 Evaluation Metrics

To evaluate the accuracy of rating prediction, we use two standard metrics: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Additionally, we employ BLEURT [39] and GPTScore [45] to assess the semantic alignment between the diverse user belief y and the target item reviews. BLEURT utilizes contextual embeddings for similarity measurement, while GPTScore provides a rating scale of 0–100 based on GPT-4o analysis.

4.3 Baselines

To comprehensively evaluate the performance, we compare SCoTRec with three types of recommendation methods: (1) traditional recommendation models (MF[22], DeepCoNN [60], DAML[27]), (2) LLM-based methods (fine-tuned Llama-3.1-8B-Instruct [16], GPT-4o [30]), and (3) CoT-based methods (Rec-SAVER [44], MAPLE [51], EXP3RT [18]). Two variants of the model, SCoTRec-LLaMA and SCoTRec-GPT-4o, are also provided.

- **MF [22]** is a representative latent factor-based CF method.
- **DeepCoNN [60]** utilizes graph neural networks to directly learn deep representations from raw review text.
- **DAML [27]** integrates rating features and review features into a unified neural network model and realizes high-order nonlinear interaction of features through a neural factorization machine.
- **Llama-3.1-8B-Instruct [16]** uses fine-tuned Llama-3.1-8B-Instruct to directly predict ratings based on the user’s rating history.
- **GPT-4o [30]** uses GPT-4o to directly predict ratings based on the user’s rating history.
- **Rec-SAVER [44]** employs CoT reasoning to analyze user history and preferences, building complex and dynamic user representations for alignment.
- **MAPLE [51]** achieves personalized recommendations by extracting reviews through LLM
- **EXP3RT [18]** constructs user and item profiles from historical reviews and ratings using LLMs. These profiles are then leveraged to reason about user preferences.
- **SCoTRec-LLaMA** adopts Llama-3.1-8B-Instruct as the sentiment reward model.
- **SCoTRec-GPT-4o** adopts GPT-4o as the sentiment reward model.

Table 1: Statistics of the Datasets

Dataset	# Users	# Items	# Interactions	# Social Ties
Epinions	14,680	233,261	447,312	632,144
Yelp	99,262	105,142	672,513	1,298,522
Music	4,183	2,660	47,638	-
Book	13,863	13,515	83,657	-

4.4 Implementation Details.

All experiments are conducted on an NVIDIA A6000 GPU with 48 GB of memory. We explore CoT N values of $\{1, 2, 3\}$, reasoning results M values of $\{6, 8, 10, 12\}$, and threshold ε values from $\{1, 2, \dots, 9\}$. Fine-tuning is performed using QLoRA on the llama-3.1-8B-Instruct model with LoRA rank 16, alpha 32, and dropout 0.1, for 3 epochs, a sequence length of 4096, and batch size 32, using the AdamW optimizer (learning rate $3e-4$, 100 warmup steps). A fixed random seed of 42 ensures reproducibility. For each user, we sample five first-order neighbors for social context, label 200 reviews, and generate a 5-item review sequence. The social CoT reasoning aggregation layer is set to 2 layers.

4.5 Overall Performance Analysis

In this experiment, we compared the performance of different models across four datasets (Epinions, Yelp, Music, and Book) using MAE and RMSE as evaluation metrics, as shown in Table 2. The results demonstrate that the SCoTRec series of models, particularly SCoTRec-LLaMA, exhibit a significant advantage in both MAE and RMSE. Compared to traditional recommendation methods, LLM-based approaches perform well in certain scenarios, due to the extensive world knowledge and strong reasoning capabilities of LLMs. This enables them to better reason user ratings for items based on reviews, leading to more accurate recommendations. Fine-tuned LLaMA outperforms GPT-4o on specific datasets (e.g., Yelp and Book), as the fine-tuned model is more suited to particular tasks. Additionally, the CoT approach has made further progress by designing a process that includes more necessary factors, enabling more complex reasoning. The EXP3RT method, a relatively effective model, achieves thoughtful reasoning through multiple steps—preference extraction, matching, and prediction. However, the reasoning unit in these methods is typically a single individual, which can lead to LLMs displaying ingratiating behavior and exacerbating confirmation bias. This feedback loop prioritizes information that aligns with the model’s existing opinions, leading users to overestimate the popularity of their views.

SCoTRec enhances recommendation performance by modeling user sentiment and incorporating a social CoT mechanism. This design leverages users’ social networks to conduct fine-grained reasoning across different sentiment pathways, thereby mitigating confirmation bias. Among its variants, SCoTRec-LLaMA outperforms others, as LLaMA’s fine-tuning enables a more precise perception of sentiment within specific task datasets. LLaMA demonstrates superior ability in capturing and interpreting sentiment information. In contrast, although GPT-4o is a powerful pre-trained model that

Table 2: Performance comparison on four datasets in terms of MAE and RMSE. The best results are shown in bold, and sub-optimal results are underlined. The SCoTRec-llama3.1-8B and SCoTRec-GPT-4o are variants of our SCoTRec model.

Method	Epinions		Yelp		Music		Book	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
MF	0.846	1.086	0.798	1.0711	0.6188	0.8142	0.6277	0.8565
DAML	1.162	1.383	0.7964	<u>1.0405</u>	0.5703	<u>0.7848</u>	0.6214	<u>0.8371</u>
DeepCoNN	0.7959	1.0392	0.8177	1.0785	0.6034	0.8057	0.6221	0.8403
LLaMA	0.6643	0.8726	0.8586	1.126	0.5775	0.8017	0.6415	0.9528
GPT-4o	0.6294	0.7764	0.8766	1.3005	0.7438	1.1069	0.7591	1.1558
Rec-SAVER	0.5986	0.7014	<u>0.8295</u>	1.1282	0.6463	0.9262	0.6645	0.9356
MAPLE	<u>0.5694</u>	<u>0.6896</u>	0.8543	1.3156	0.6637	0.9654	0.6854	0.9648
EXP3RT	0.5781	0.6897	0.8306	1.2311	<u>0.5608</u>	0.8385	<u>0.6135</u>	0.9370
SCoTRec-LLaMA	0.5333	0.6479	0.7141	0.9827	0.5253	0.7589	0.5793	0.8326
SCoTRec-GPT-4o	0.5337	0.6474	0.7346	1.027	0.5326	0.7530	0.5799	0.8531

Table 3: Quality evaluation of generated text. The best results are highlighted in bold, and sub-optimal results are underlined. The Gscore denotes the GPTScore metric, and BLE denotes the BLEURT metric.

Method	Epinions		Yelp		Music		Book	
	Gscore	BLE	Gscore	BLE	Gscore	BLE	Gscore	BLE
Rec-SAVER	74.58	0.4276	<u>66.43</u>	0.4102	75.60	0.3652	72.45	0.4233
MAPLE	<u>77.42</u>	<u>0.4563</u>	62.58	0.4036	73.96	0.3763	71.69	0.4196
EXP3RT	76.42	0.4436	64.28	<u>0.4275</u>	<u>76.22</u>	<u>0.3840</u>	<u>73.60</u>	<u>0.4373</u>
SCoTRec	88.56	0.4932	88.39	0.5128	89.72	0.5069	89.49	0.5375

performs well on general tasks, the fine-tuned LLaMA in our recommendation case exhibits stronger sentiment and contextual understanding, effectively integrating sentiment signals with user-item rating relationships to improve recommendation performance.

Next, we analyze the reasoning quality of SCoTRec, focusing on the results from the GPTScore and BLEURT metrics across four different datasets, as shown in Table 3. The results show that SCoTRec outperforms all other methods on both the GPTScore and BLEURT metrics across all datasets, achieving significant improvements compared to the others. This superiority can be attributed to SCoTRec’s effective integration of user sentiment, and its ability to leverage diverse sentiment feedback from users’ social networks, which helps alleviate confirmation bias and thereby enhances the quality of the reasoning.

4.6 Ablation Study

The ablation study investigates the impact of different components on the performance of SCoTRec. We include five ablation variants: ‘w/o SCoT’, indicating the removal of the social CoT, replaced by single chain reasoning; ‘w/o SN’, meaning that social network information is omitted, and users interacting with the same item are treated as having potential social relationships; ‘w/o UPC’, which refers to the exclusion of user profile construction, relying instead

Table 4: The ablation experiment results. The best results are shown in bold.

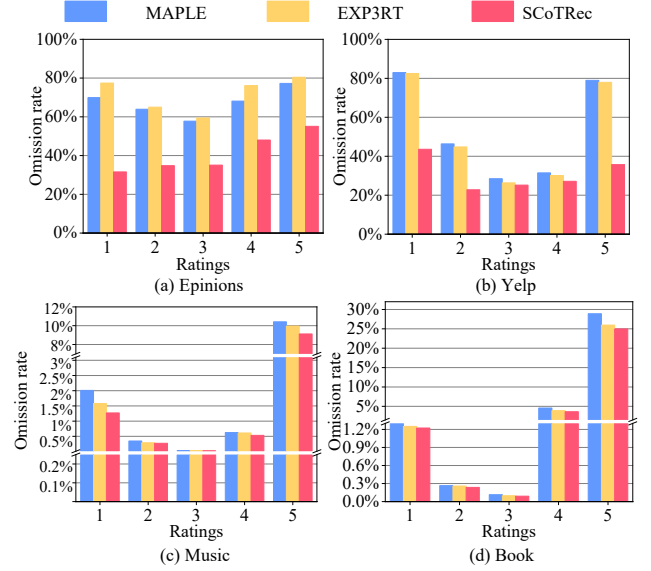
Method	Epinions		Yelp		Music		Book	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
-w/o SCoT	0.5467	0.6611	0.7596	1.172	0.5368	0.7583	0.6005	0.8742
-w/o SN	0.5443	0.6591	0.7525	1.1712	-	-	-	-
-w/o UPC	0.5837	0.7027	0.7756	1.2846	0.539	0.7638	0.6164	0.8885
-w/o MRR	0.5458	0.6615	0.7438	1.0569	0.5376	0.7602	0.6127	0.8782
-w/o CE	0.5578	0.6879	0.7596	1.2163	0.5387	0.7654	0.6156	0.8859
SCoTRec	0.5333	0.6479	0.7141	0.9827	0.5253	0.7589	0.5793	0.8326

on user interaction records and review information to build the user profile; ‘w/o MRR’, which means the omission of multiple reasoning result, using only a single reasoning result; ‘w/o CE’, which indicates that the consistency evaluation score function is not used and reasoning and aggregation is performed directly.

The results of our ablation study across four datasets in terms of MAE and RMSE metrics are shown in Table 4. Based on the experimental results, we found that removing each component led to a decline in model performance, further confirming the critical role of each important component. Without social CoT, the “w/o SCoT” model fails to integrate diverse user beliefs, causing the LLM to suffer from certainty bias and resulting in degraded recommendation performance. ‘w/o SN’ ignores the real social relationships between users causes the model to lose some social context, thus affecting the prediction of user preferences. ‘w/o UPC’ results in the worst performance, indicating that without sentiment awareness, the model relies solely on user interaction records and review information to construct user profiles, leading to a significant increase in both MAE and RMSE. This suggests that sentiment awareness is crucial for understanding user preferences and providing accurate recommendations. ‘w/o MRR’ shows that a single reasoning result cannot provide effective recommendations compared to using multiple reasoning results. ‘w/o CE’ leads to the inability to evaluate the reasoning result, and directly performing aggregation affects the quality of the reasoning results.

4.7 Confirmation Bias Analysis

Confirmation bias occurs when users are primarily exposed to content that aligns with their existing preferences and views, reinforcing rather than challenging their true opinions. In this section, we conduct the experiment aiming to evaluate whether SCoTRec can alleviate confirmation bias in RSs to some extent. We use the measure of omission rates of positive and negative traits at different rating levels (ranging from 1 to 5) to reflect relative confirmation bias, and compare SCoTRec with the baseline models, including the MAPLE and EXP3RT, on the four datasets. MAPLE uses a pre-trained model based on T5 [34] to directly extract sentiment tuples from user reviews for explanation generation. EXP3RT infers a set of user preferences (likes and dislikes) from reviews. Experimental results in Figure 3 show that SCoTRec reduces the omission rate of sentiment traits compared to MAPLE and EXP3RT, with particularly significant improvements on the social datasets Epinions and Yelp. This highlights the effectiveness of leveraging the social CoT

**Figure 3: The omission rate results of sentiment traits on four datasets.**

to integrate diverse user perspectives within the social network for complex reasoning to some extent. Overall, SCoTRec encourages users to explore a variety of positive and negative emotional traits and reduces the reinforcement of existing opinions, effectively alleviating confirmation bias.

4.8 Hyperparameter Analysis

The Number of CoT. Table 5 reports results for different CoT step values. With one step, the model unifies positive and negative reasoning; with two, it separates positive–positive and negative–negative chains. Performance generally improves as the number of steps increases, peaking at three. An exception occurs on the Music dataset, where two steps yield the best RMSE, likely due to its smaller size—separating positive and negative reasoning better captures sentiment relations and reduces prediction errors.

The Number of Reasoning Results. Table 6 presents the results for different numbers of reasoning units. The findings indicate that as the number of reasoning results increases, the model is able to better handle and analyze the complex relationships between users and items, leading to more accurate predictions. However, when the number of reasoning results reaches a certain threshold, model performance begins to decline, indicating the occurrence of over-reasoning. In the Epinions, Yelp, and Book datasets, the model performs best when the number of reasoning results is set to 10. In contrast, on the Music dataset, MAE performs best with 10 results, while RMSE performs best with 8 results.

The Threshold of ϵ . Table 7 presents the results for different sizes of ϵ . Performance improves as ϵ increases from 1 to 5, indicating that overly permissive thresholds (small ϵ) allow low-quality reasoning results to be retained, harming accuracy. The best results are generally achieved around $\epsilon = 5$ to 7, balancing selectivity and reasoning diversity. When ϵ becomes too large (e.g., 9), performance slightly

Table 5: Performance comparison of the number of CoT across four datasets. The best results are in bold.

# CoT	Epinions		Yelp		Music		Book	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
1	0.5467	0.6611	0.7596	1.172	0.5368	0.7583	0.6005	0.8742
2	0.5414	0.6560	0.7326	1.0942	0.5348	0.7536	0.6105	0.8842
3	0.5333	0.6479	0.7141	0.9827	0.5253	0.7589	0.5793	0.8326

Table 6: The number of reasoning results performance comparison across four datasets. The best results are bolded.

# reasoning	Epinions		Yelp		Music		Book	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
6	0.5381	0.6534	0.7179	0.9894	0.5462	0.776	0.5864	0.8699
8	0.5317	0.6468	0.7141	0.9827	0.5278	0.7593	0.5823	0.8579
10	0.5333	0.6479	0.7150	0.9806	0.5253	0.7589	0.5793	0.8326
12	0.5428	0.6572	0.7277	0.9963	0.5487	0.7791	0.5808	0.8465

Table 7: The size of ϵ performance comparison across four datasets. The best results are bolded.

Value of ϵ	Epinions		Yelp		Music		Book	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
1	0.6842	0.8392	0.8942	1.3962	0.6356	0.8764	0.6942	1.0369
2	0.6543	0.7658	0.8796	1.2960	0.5847	0.8642	0.6756	0.9652
5	0.5967	0.7265	0.8146	1.0563	0.5253	0.7589	0.5793	0.8326
7	0.5333	0.6479	0.7141	0.9827	0.5648	0.7892	0.6293	0.8756
9	0.5694	0.6052	0.7642	1.0363	0.5792	0.8142	0.6576	0.9231

degrades, likely due to excessive discarding of useful reasoning results.

4.9 Case Study

To further examine whether SCoTRec’s reasoning results can mitigate confirmation bias, we conducted a case study, as shown in Figure 8. Based on the user’s review and rating, it can be inferred that the user holds certain negative opinions about Baru, though such details cannot be directly derived from the review text. We compared the reasoning outputs of EXP3RT and SCoTRec. The results show that EXP3RT, relying solely on user information, extracted only positive aspects and predicted a rating of 5. In contrast, SCoTRec incorporates the user’s social network information, effectively capturing both positive and negative sentiment traits toward Baru and accounting for subjective sentiment, thereby producing a more accurate prediction of rating 4.

5 Related Work

Review-based Recommendation. Review-based recommendation [7, 28, 29, 41] typically integrates the rich information contained in reviews into the modeling of users and items. BoWLF [1] utilizes a distributed bag-of-words representation of reviews to regularize

the latent factors in a matrix factorization approach for collaborative filtering. Later studies, such as DeepCoNN [60] and DAML [27], begin to utilize graph neural networks to directly learn deep representations from raw review texts. RRM [59] incorporates social network information and sentiment tendencies from reviews into the recommendation task, effectively improving rating prediction accuracy. However, these review-based recommendation methods often suffer from the cold-start problem. Different from them, our SCoTRec leverages the rich world knowledge of LLMs to extract aspect-level sentiment terms from limited reviews, thereby alleviating the cold-start problem.

LLM-based Recommendation. LLM4Rec [4, 6, 11, 26, 33, 40, 54, 61] leverages the world knowledge and reasoning capabilities of LLMs to address the cold-start problem in RSs. While methods like CoLLM [57] use traditional models to integrate user behaviors into LLMs, the semantic gap between LLMs and RSs poses a significant challenge to effective knowledge transfer. To address this, PrefRec [50] introduces an RLHF framework to align user behaviors with LLMs, despite the high computational cost of reinforcement learning. To further reduce the reliance on reinforcement learning, role-playing approaches are then proposed to align LLMs with user behaviors through user profiles. For example, ExpertLLaMA [49] uses expert profiles and real-world instructions to guide LLMs to generate high-quality responses, while Rec-SAVER [44] employs CoT reasoning to analyze user history and preferences, building complex and dynamic user representations for alignment. However, these models often suffer from confirmation bias. In comparison, SCoTRec effectively alleviates confirmation bias by incorporating sentiment feedback from social users into the reasoning unit.

6 Conclusion and Discussion

In this study, we propose SCoTRec, a recommendation method based on social CoT LLM reasoning that explores the social CoT in the LLM-based recommendation. By primarily leveraging a social CoT mechanism with sentiment-based pathways and simulating human evaluation operations, SCoTRec effectively alleviates confirmation bias.

In this paper, SCoTRec is built on social networks. When such information is unavailable, we adopt an alternative approach by treating users who interacted with the same item as potential social links (Music and Book datasets). Although SCoTRec still outperforms baseline models on these datasets, the improvements are smaller than on Epinions and Yelp, suggesting that this alternative may introduce additional noise. In future work, we aim to improve the design of CoT applications to tolerate noise in recommendations. Similar to GRSU [46], generating multiple reasoning results in SCoTRec may increase computation.

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A Appendix

A.1 Case study

Table 8: The blue text denotes negative aspects and red text denotes positive aspects.

Review & Rating
I feel that Baru is underrated. Not many people recommend it. However, I've had really delicious entrees there. I do feel that they could do better or just be more consistent with their drinks. I think one of my favorite nights on magazine ended with my husband and in laws on the baru balcony... User Ratings: 4.
EXP3RT
The user would appreciate the underrated charm and welcoming atmosphere of Baru, as they value hidden gems that offer memorable experiences. The cozy balcony dinner with family reflects their preference for warm, story-like moments that create lasting impressions. Their attention to both the quality of entrees and the inconsistency of drinks shows an appreciation for detail and coherence, similar to enjoying narratives with strong world-building and logical flow. Overall, the user would likely enjoy Baru for its inviting setting, flavorful dishes, and the sense of discovery that makes the visit feel special. User rating prediction results: 5.
SCoTRec
The user's preferences and review history suggest a likely mixed experience at Baru Bistro & Tapas. While the user values high-quality food, a pleasant dining experience, and good service—elements that are positively highlighted in the reviews of Baru Bistro & Tapas—the negative aspects, particularly regarding drink quality and service, could overshadow these positives.... Therefore, while the food quality and dessert may appeal to the user, the shortcomings in drink strength and service quality may lead to disappointment, suggesting that ... User rating prediction results: 4.

A.2 Aspect-level Sentiment Terms Prompt

Aspect-level sentiment terms prompt
<Instruction> Based on the given example, the user's comment history for each item is divided into two parts, positive and negative. Please strictly adhere to the response format in the example without any other sentences. <User Review History> {user_review} <Example> Input: <User Review History> I absolutely love these wireless Bluetooth headphones. They are incredibly lightweight and comfortable to wear, with a long battery life. The sound quality is clear, with deep bass and crisp highs. However, the charging case is prone to scratches, and sometimes the connection is unstable. Overall, I'm very satisfied; they are worth the price. Output: Positive Aspects: Comfortable, Lightweight. Negative Aspects: Scratch-Prone Case, Unstable Connection.